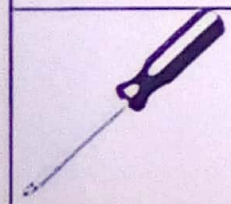
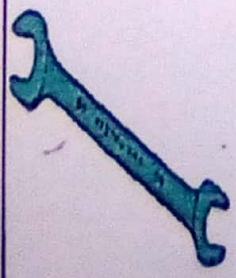
TOOLS IDENTIFICATION

			
A: FLAT screw	B: TORX screw	C: STAR screw head	D: ALLEN screw head

			
1 CLAW HAMMER	2 BALL PANE HAMMER	3: HACKSAW	4: LONG NOSE PLIER
			
5: STAR SCREW DRIVER	6: FLAT SCREW DRIVER	7: TORX DRIVER	8: ALLEN KEY
			
9: SET OF HERMAPHRODITE SPANNERS	10: FLAT SPANNER	11: PIPE WRENCH SPANNER	12: ADJUSTABLE SPANNER

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KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

B.Sc. (Engineering) First Semester Examination, 2012, First Year

ME 195/CE 153: ENGINEERING TECHNOLOGY 1 **TIME ALLOWED: 2½3 Hrs**

ANSWER ALL QUESTIONS. In PARTS I, II & III SEPARATELY.

NO BOOKS ARE ALLOWED ***INDEX NUMBER:*** **……**

PART I: MECHANICAL **DEPARTMENT:……..**

Q1a Sketch below the Steam Boiler at the Thermodynamics Laboratory, KNUST and label ONLY the safety devices *(12marks)*.



Q1b State four (4) applications of Steam Boiler *(8marks)*:

i. _____

ii. _____

iii. _____

iv. _____

Q5 Underline the correct statement that best describes the **following major parts in an automobile engine either as an internal or external part.** *(12marks)*

Radiator	**Internal part/External part**
Carburettor	**Internal part / External part**
Piston	**Internal part/External part**
Flywheel	**Internal part / External part**
Battery	**Internal part / External part**
Crankshaft	**Internal part / External part**
Cam Shaft	**Internal part / External part**
Manifold	**Internal part / External part**
Connecting Rod	**Internal part / External part**
Gudgeon Pin	**Internal part / External part**
Timing chain	**Internal part / External part**
Spark plug	**Internal part / External part**

Q6a **CHOOSE THE CORRECT WORD/SET:** *(6marks)*

A car uses a four-stroke engine. The four strokes are

intake, compression, power and exhaust

injection, rotation, power and exhaust

injection, carburetion, rotation and exhaust

b State two (2) main types of automobile engine cooling system *(4marks)*

(i) _____ (ii) _____

Q7. What machine tool is best suited for producing each of the following? (*12 marks*)

a) gear teeth _____

b) external keyway _____...

c) internal keyway. _____

d) multiple holes in a flat plate _____.

e) smooth flat surface _____

f) thread on a short shaft. _____

Q8. List any three types of fire extinguishers (*6 marks*)

I _____

Ii _____

iii _____

Q9. Suggest **four** uses of compressed air (*4 marks*)

i._____ ii. _____

iii. _____ iv. _____

Q10. Name any three other alternate sources of non -conventional energy (*6 marks*)

i. _____

ii._____

iii. _____

Q11. Sketch and label a carburizing flame in the box (*5marks*)

Q12. Sketch and label a neutral flame in the box *(5marks)*

Q13. Sketch and label an oxidizing flame in the box (*5 marks*)

Q14. Sketch and fully label a single-core PVC-insulated PVC-covered cable. (*5 marks*)

Q15. State the reading on the micrometer shown in figure 2 (*5 marks*)

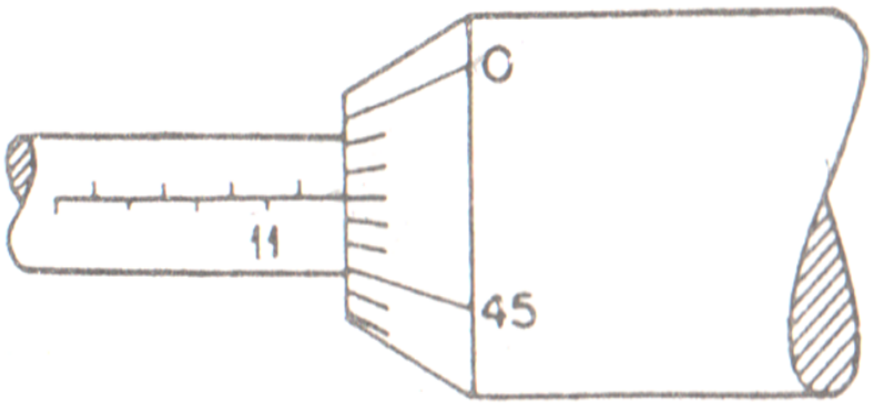


Figure 2

Ans: .....

Q16. Which state institution is responsible for the compliance of the Industrial SafetyAct (*2 marks*)

_____

Q17. Figure 3 is a sketch developed by a student for his/her engineering technology report. State the name of the equipment and label the parts A – K. _____ *(15marks)*

A _____

B _____

C _____

D _____

E _____

F _____

G _____

H _____

I _____

J _____

K _____

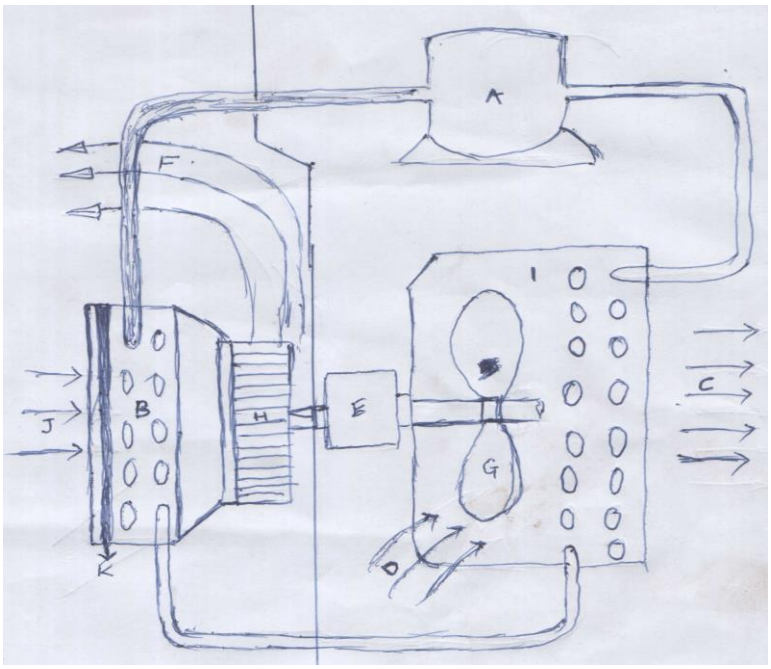


Figure 3

Q18. State the names of the labels A – D of gas welding set as shown in Figure 4 (*8 marks*)

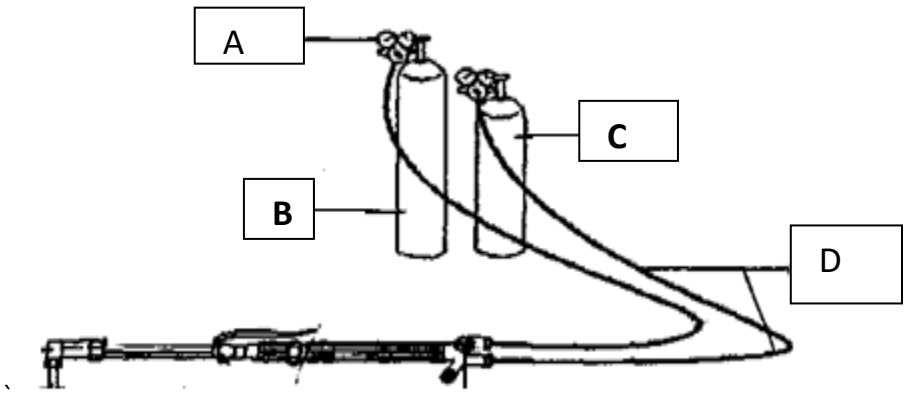


Figure 4: Gas welding set

A …… B ……..

C……... D……..

Q19 State the difference between a Refrigerator and an Air Conditioner *(4marks)*

_____

_____

_____

Q20a List the equipment you will need to provide solar PV electricity to power LCD projector in a classroom . (*10 marks*)

_____ _____

_____ _____

_____ _____

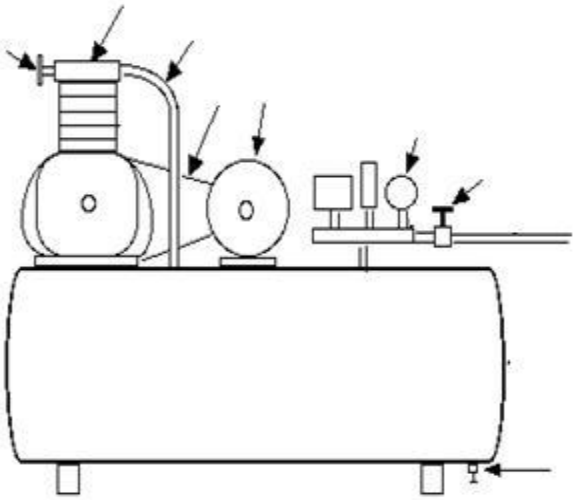
b. Show in the box below how the solar PV equipment you have listed in Q20a can be connected together. *(10marks)*

Q21. Write the name of the instrument shown in figure 5 (2 *marks*)



***Figure 5*_____**

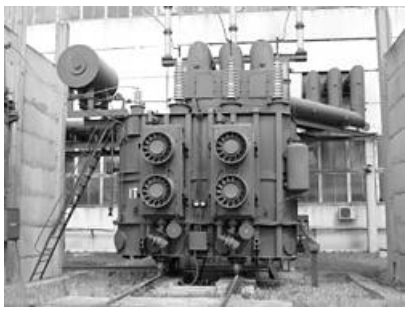
Q22. Label the diagram shown below. What equipment is this? _____ *(20marks)*



Q23 Electricity is produced at power plants and it has to get to the customers that use the electricity through conductors and other devices. State the names of the electrical devices and equipment below. *(15marks)*







A _____ **B.**_____ **C** _____

Q24. List the order in which the following electrical components A-G will be connected for electricity generated to be supplied to a household: *(10marks)*

A. Transmission Line **B**. Consumer Control Unit (CCU) **C.** Distribution Line

D. Transformer Sub-Station **E**. Cut Out **F** Meter **G**. Service Cable

1 _____ *2.* _____

3. _____ *4.* _____

5. _____ *6.* _____

7. _____

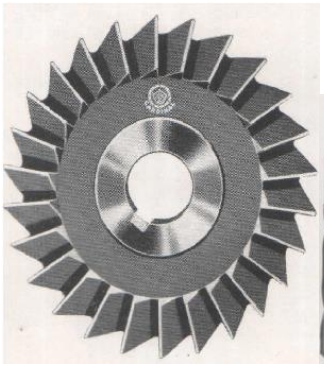
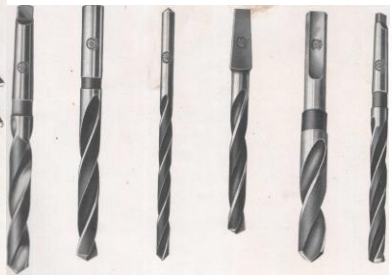
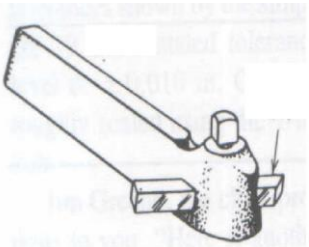
Q25. Indicate the voltages at which electricity from the National Grid in Ghana is distributed: *(10marks)*

i. Transmission lines _____ V ii. High Voltage lines _____ V

iii. Low Voltage lines _____ V iv Three Phase Supply _____ V

v Single Phase Supply _____ V

Q26. Name the cutting tools shown in the figures below (*20 marks*)

*A:*_____*B:*_____ *C:*_____*D:* _____

Q27. List at least two (2) machine tools that employs or uses the cutting tools as shown in Q26 (*8 marks*)

	A	B	C	D
1				
2				

Q28. Name two (2) types of work holding devices on the centre lathe machine (*4 marks)*

i. _____ ii. _____

Q29. Name the machine tool in the Mechanical Engineering workshop that uses magnetic vice (*2 marks)* _____

Q30. Which part of the radial drilling machine houses the driving mechanism? (*2marks)*
_____

Q31a. Two (2) shafts on the centre lathe machine control the movement of the carriage. State the names of these shafts. (*4 marks)* i _____ii. _____

b. Which of the two (2) shafts stated in question **Q31a** is engaged in screw cutting process? (*2 marks)*_____

Q32. Name any three (3) accessories used in electric arc welding and state their uses. (*6marks)*

A._____

B._____

C. _____

Q33. State three (3) functions of flux used in gas welding process. (*6 marks)*

i _____

ii. _____

iii. _____

Q34. Explain the difference between electric arc welding and Oxy-acetylene (*4 marks)*

_____

_____

_____

_____

_____

_____

_____

Q35 Sketch the chassis in the automobile workshop (KNUST) showing the components of the transmission in the space provided below and label the parts. (*20 marks*)

NKAY | NANA KWABENA OSEI #SimplyCharismatic

Q36 Underline the instruments that are NOT capable of giving direct readings: (*5 marks)*

i. Pair of dividers
ii. Vernier depth guage
iii. Thread guage
iv. Telescope guage
v. Steel Rule

Q37 State the name of the machine tool in figure 5 (*2 marks)*

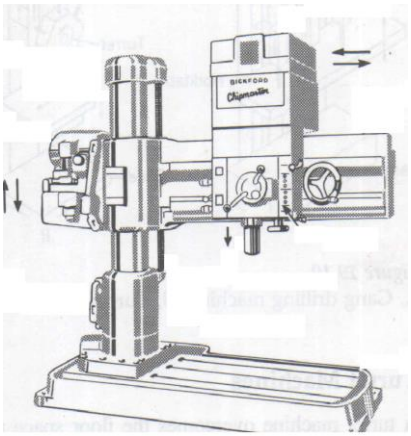


Figure 5: _____

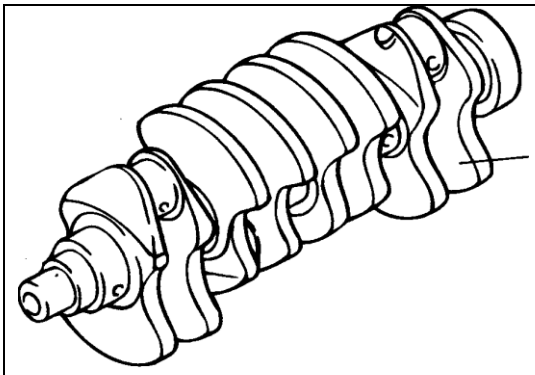
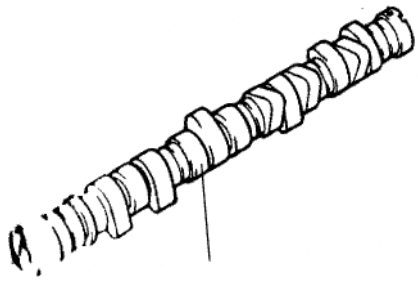
Q38 State any <u>FOUR</u> applications of solar energy *(8 marks)*

i. _____
ii. _____
iii. _____
iv. _____

Q39 Sketch and fully label a single-core PVC-insulated steel wire armoured PVC-covered (PVC/SWA/PVC) cable. (*5 marks*)

NKAY | NANA KWABENA OSEI
#SimplyCharismatic

Q40 Write down the appropiate names of the automobile engine components as shown in figure below *(4 marks)*

	
A:	B:

NKAY | NANA KWABENA OSEI #SimplyCharismatic

I. A. Edwin

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING END OF SEMESTER EXAMINATIONS

B.Sc. (Engineering) First Semester Examination, 2009/10. First Year

ME 195/CE153: ENGINEERING TECHNOLOGY 1 *PARTI* TIME

ALLOWED: 3hrs

NO BOOKS ARE ALLOWED ANSWER ALL QUESTIONS.

Q1 . a) Describe how a 4-cylinder petrol engine works and state the two possible firing orders.

b) State two main differences between a petrol engine and a diesel engine.

c) Briefly describe how power transmission from the engine to the drive wheels.

Q2. a) With the aid of a diagram, describe the basic gas welding process.

b) Sketch and label the electric arc welding equipment.

Q3. a) Which machine tools will you select for the following operations?

i) Produce a shaft with steps and a thread at one end and keyway at the other end. (two machines)

ii) Produce a gear with a small bore and a keyway. (three machines)

iii) Produce one thousand bolts and nuts, (one machine)

b) List four (4) measuring instruments excluding the rule, Sketch any two of them and show how they are used to measure a specific value.

Q4. a) Explain briefly how solar energy is used to operate a 240V/10A refrigerator.

b) Name any three other alternate sources of non -conventional energy.

c) Name two types of refrigeration compressors and explain two principal differences.

d) With the aid of a diagram, describe how an air conditioner works.

Q5. a) Explain how a horizontal oil-fired fire tube boiler generates superheated steam. b)

State three uses/applications each of the following?

i. Steam ii) Compressed air

Agbeko, E. E, K. Edwin, I. A. Idan, C. K.

In a refrigerator, the..... compresses the.....

gas. 1 his raises the ..... pressure and.....so the heat-exchanging coils outside the refrigerator allow the .....to dissipate the heat of pressurization.

As it cools, the refrigerant .....into liquid form and flous through the .....valve. When it flows through the ..... valve, the liquid refrigerant is allowed to move from a .....pressure zone to a ..... pressure zone, so it expands and .....•..... In so doing, it absorbs....., ... making it cold. The coils inside the refrigerator allow the ..... to absorb..... making the inside of the refrigerator cold. The cycle then repeats.

Carnage .....

Bed.....

Dead Centre .....

Longitudinal and Transverse Feed Control .....

Face plate .....

Q3. Insert the name of the appropriate machine in the following sentences:

A .....Machine is generally/commonly used to manufacture gear teeth

A ..... Machine is generally/commonly used to manufacture external keyways

A ..... Machine is generally/commonly used to manufacture internal keyways

A ..... Machine is generally/commonly used to make multiple holes in a flat plate

A ..... Machine is generally/commonly used to make smooth flat surfaces

A .....Machine is generally/commonly used to manufacture one million bolts

A ..... Machine is generally/commonly used to manufacture thread on a short shaft

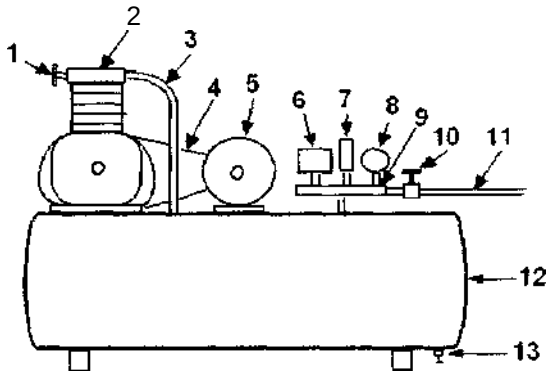
Q4. Label the electric arc welding schematic shown below

List any five types of fire extinguishers

IV. **State five ways of providing adequate protective disease control measures**

IV.

Q9. LABEL THE DIAGRAM SHOWN BELOW. What equipment is this?



Q1. State one function of the following parts in an automobile.
Radiator ……
Carburetor ……
Piston rings ……
Piston ……
Flywheel ……
Alternator ……
Battery ……..
Condenser ……
Crankshaft ……
Ignition coil ……
Gearbox ……
Propeller Shaft ……
Clutch ……

Contact Set ……
Cam Shaft ……
Muffler/Silencer ……

CHOOSE THE CORRECT WORD/SET:
A car engine converts fuel into: i: exhaust gases ii: heat iii: motion

A car uses a four-stroke engine. The four strokes are
intake, compression, ignition and exhaust
injection, rotation, ignition and exhaust
injection, carburetion, rotation and exhaust
A turbocharger is:
a set of gears that makes the wheels turn faster
a turbine that compresses the air traveling into the engine
an injector that delivers fuel to the engine faster
List three most common type of springs used in modern car suspensions?
i…… ii…… iii……
Which type is used in the chassis in the automobile workshop? ……
Q2. State one function of the following parts in a Lathe machine.
Apron ……
Split Nut Control……
Clutch ……
Feed Selector ……
Hand Wheel ……
Spindle Speed Selector ……
Chip Pan ……
Cross Slide ……
Compound ……
Headstock Assembly ……..
Tool Post ……
Guide ways ……
Spindle with Chuck ……
Tail Stock Assembly ……
Feed Rod (Screw) ……
Lead Screw ……

Carriage …….

Bed ……..

Dead Centre ……

Longitudinal and Transverse Feed Control …….

Face plate ……

Q3. Insert the name of the appropriate machine in the following sentences:

A …… Machine is generally/commonly used to manufacture gear teeth

A ……..Machine is generally/commonly used to manufacture external keyways

A …… Machine is generally/commonly used to manufacture internal keyways

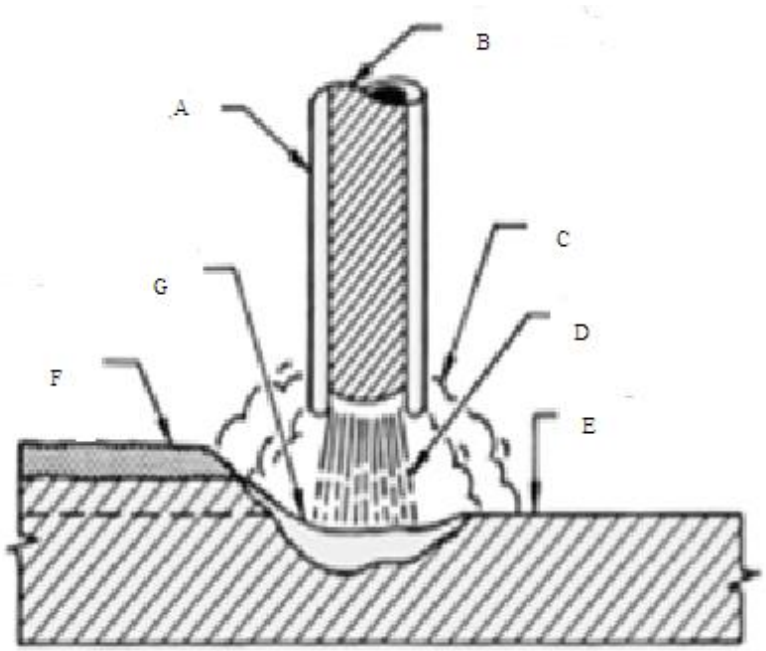
A …… Machine is generally/commonly used to make multiple holes in a flat plate

A …… Machine is generally/commonly used to make smooth flat surfaces

A …… Machine is generally/commonly used to manufacture one million bolts

A …… Machine is generally/commonly used to manufacture thread on a short shaft

Q4. Label the electric arc welding schematic shown below



A…….. B……. C…….

D…….. E……. F…….. G……

Fill in the blank spaces with the appropriate word:

In arc welding, the intense …….. needed to melt metal is produced by an …….. arc. The arc is formed between the actual work and an …… that is manually or mechanically guided along the joint. Arc welding may be done with ……. current with the electrode either positive or negative or …… ……. current. The choice of current and polarity depends on the ……, the type of ……, the arc atmosphere, and the ……. being welded

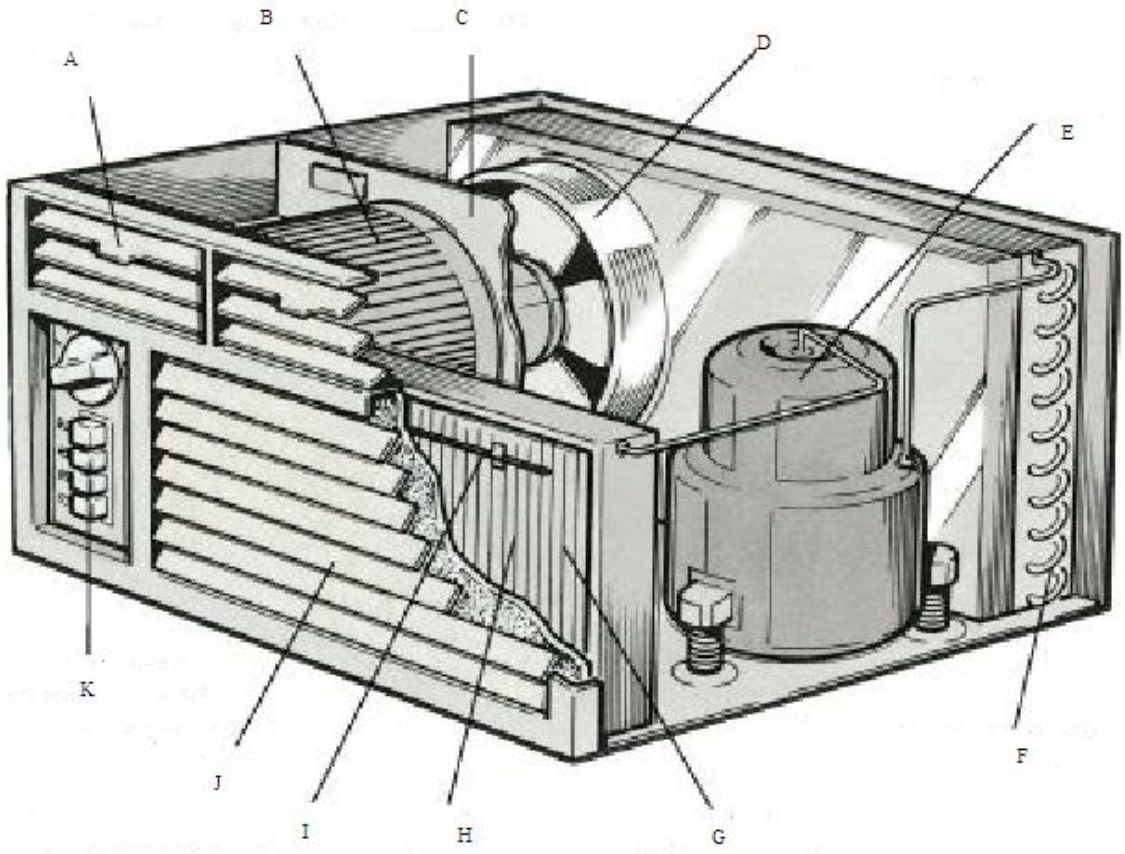
Q5. Fill in the blank spaces with the appropriate word

In a refrigerator, the …….. compresses the ……... gas. This raises the …… pressure and …….., so the heat-exchanging coils outside the refrigerator allow the …….. to dissipate the heat of pressurization.

As it cools, the refrigerant ……..... into liquid form and flows through the …… valve. When it flows through the …….. valve, the liquid refrigerant is allowed to move from a

…….. pressure zone to a ……. pressure zone, so it expands and ……... In so doing, it absorbs ……..…..... making it cold. The coils inside the refrigerator allow the …….. to absorb ……, making the inside of the refrigerator cold. The cycle then repeats.

LABEL THE DIAGRAM SHOWN BELOW



A…….. B……. C…….

D…….. E……. F……..

G…….. H……. I…….

J…… K……..

What are the functions of:

B……...

D…….

E…….

F……..

G……..

Q6. State the instruments you would use to measure the following:

An internal measurement (bore) to an accuracy of two decimal places ……...

An internal measurement (bore) to an accuracy of three decimal places …….

The depth of a hole estimated to be about 51 mm to two decimal places ……

The depth of a hole estimated to be about 51 mm to three decimal places ……

A round solid bar 55.355 mm. in diameter …….

The included angle of a taper gauge……

Checking a horizontal surface for leveling……..

What is meant by a standard M 46/2 slip gauge set

…….....

.....

Q7.It is required to provide 220-240 volt solar power system to one of the classrooms in the faculty. Make a list of six components and materials required and state the purpose of each one.

Component/material	Purpose
……	……
……	……
……	……
……	…….....
……	……
……	……

State any SIX applications of solar energy

i. .....

ii. .....

iii. .....

iv. .....

v. .....

vi. .....

Q8. List the three basic elements required for a fire to start:

i …… ii …… iii ……

In machines, the hazards to the operators arise from the unguarded points of operation, such as

i……

ii……..

iii ……

iv…….

In fire protection some advantages of sprinkler systems are:

i……

ii……

iii ……

iv……

v ……..……

List any five types of fire extinguishers

i……

ii……

iii ……

iv……

v ……..……

State five ways of providing adequate protective disease control measures

i……

ii……

iii ……

iv……

v ……..……

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

B.Sc. (Engineering) Supplementary Examination, 2015, First Year

ME 195/CE 153: ENGINEERING TECHNOLOGY **TIME ALLOWED: 2 hrs 30mins**

ANSWER ALL QUESTIONS. In PARTS I, II & III SEPARATELY.

NO BOOKS ARE ALLOWED ***INDEX NUMBER:*** **……**

PART I (MECHANICAL) ***PROGRAMME*/DEPARTMENT: ……..**

Q1a). Below is the SAFETY BOOT you were supplied at the beginning of this semester. State the reason why the College supplied this boot to you and all engineering students. _____

_____

The manufacturer stated six (6) features of the SAFETY BOOT as: *Toe cap; steel midsole; shock absorbent; oil resistant; petrol and chemical resistant and anti-slip* both in icons and words

b) State precisely the two (2) main features written under the safety boot _____ and _____

c) Label the safety boot below the following parts: *steel toe cap, padded tongue, padded collar, anti slip oil resistant sole, shock absorption on heels, reinforce counter and reinforced eyelets*



30 marks

Q2. State the functions of the following components associated with CENTRE LATHE machine used in the machine shop: (20 marks)

Tailstock:_____

Headstock: _____

Compound Slide: _____

Tool Holder: _____

Cross Slide: _____

Feed Shaft: _____

Lead Screw: _____

Jaws: _____

Dead Centre: _____

Carriage/Saddle/Apron: _____

Q3. A stroke is the movement of the piston from Top Dead Center (TDC) to Bottom Dead Center (BDC). What is meant by dead centers? _____

_____

_____ ***(5 marks)***

Q4 Apart from the breaking system, what other means can be used to reduce speed of a vehicle in automobile._____

_____ ***(5 marks)***

Q5. State one function for each of the following major parts in the automobile (20 marks)

a. Crankshaft _____

b. Carburetor _____

c. Piston rings _____

d. Flywheel _____

e. Alternator _____

f. Battery _____

g. Gearbox _____

h. Cam Shaft _____

Q6. Insert the name of the appropriate machine tool you saw in the machine shop of Mechanical Engineering workshop in the following sentences: ***(15 marks)***

A …… Machine is generally/commonly used to manufacture gear teeth

A ……..……Machine is generally/commonly used to manufacture external keyways

A ……..… Machine is generally/commonly used to manufacture internal keyways

A …… Machine is generally/commonly used to make multiple holes in a flat plate

A …… Machine is generally/commonly used to make smooth flat surfaces

A …… Machine is generally/commonly used to manufacture thread on a short shaft

Q7. . Name any three other alternate sources of non -conventional energy ***(6 marks)***

i.……

ii.……

iii.……

Q8. Suggest **four** uses of compressed air ***(4 marks)***

i. ……

ii. ……

iii. ……

iv……

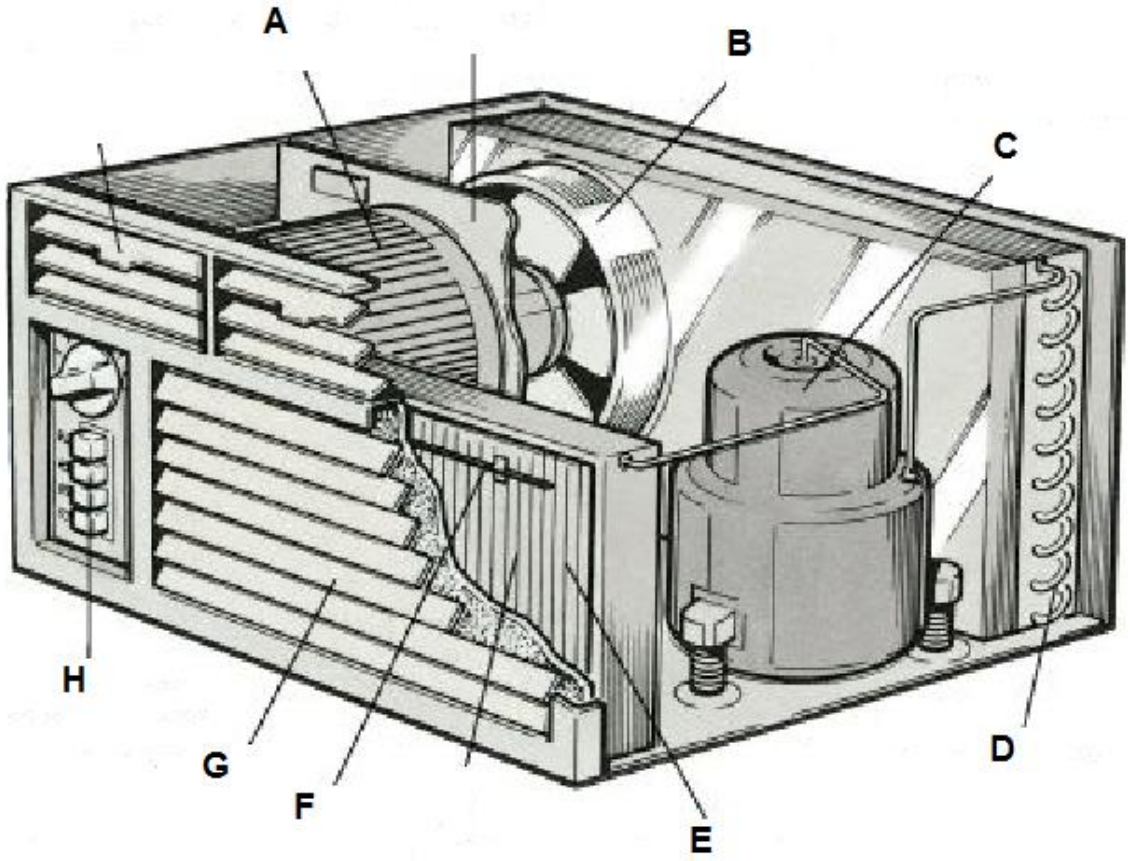
Q9. List any three types of fire extinguishers ***(4 marks)***

i. .....

ii. .....

iii. .....

Q10 LABEL THE DIAGRAM SHOWN BELOW ***(15 marks)***



A…… B…….

C……. D……..……..

E……. …… F ……..

G……. H ……...

Q11. State the reading on the micrometer shown in figure 1 ***(3 marks)***

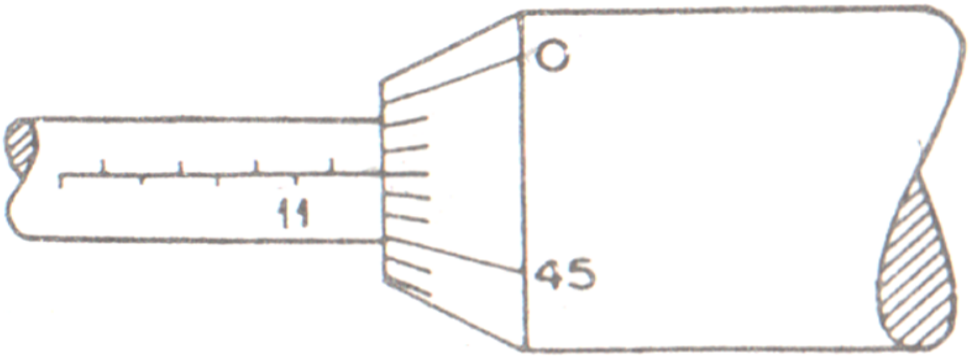


Figure 1

ANS: .....

Q12 Sketch, label and state the composition of the three (3) common flames associated with oxy-acetylene welding. ***(14 marks)***

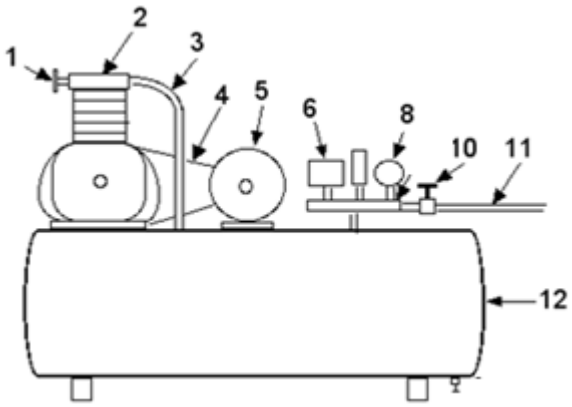
i. _____

ii. _____

iii. _____

Q13a). What equipment is shown below? ..... ***(2 marks)***

b. LABEL THE DIAGRAM SHOWN BELOW ***(10 marks)***



1	
2	
3	
4	
5	
6	
8	
10	
12	

Q14 State the institution of State responsible for the compliance of the Industrial Safety**:** ***(5 marks)***

.....

Q15 List **four** industries where Steam Boiler could be employed and state its use or application ***(4 marks)***

i. .....

ii. .....

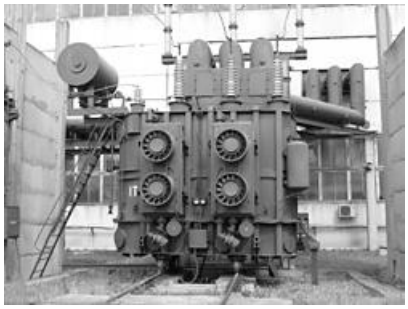
iii. .....

iv. .....

Q16. . Electricity generated at power plants get to the consumer through cables and other devices. State the names of the electrical devices and equipment listed below. ***(15 marks)***







A ..... **B.** ..... **C** .....

Q17 State the name of the machine tool in figure 2 (*2 marks)*

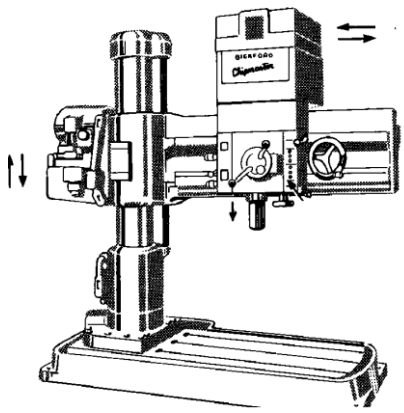


Figure 2: _____

Q18. In your room you have the following appliances and operate them on daily basis. Determine your daily energy consumption in Watt-hours (Wh), kilowatt-hours (kWh) and units of electricity (Units). ***(5 marks)***

No.	Appliance	Power (Watts)	Time (hours)	Energy (Watthours)
1	refrigerator	100	12	
2	water heater	1000	½	
3	Electric Iron	1200	½	
4	Rice cooker	900	½	
5	Television	80	6	
6	Standing Fan	70	6	
7	Laptop Computer	90	6	
8	Compact Fluorescent Lamp (CFL)	20	8	
9	LED bulb	9	10	
			Total	

i. …… Wh ii. …… kWh . iii. …… Units

Q19 Complete the wiring diagram of a Solar Home System (SHS) figure 3 below by joining the correct connecting terminals. ***(10 marks)***

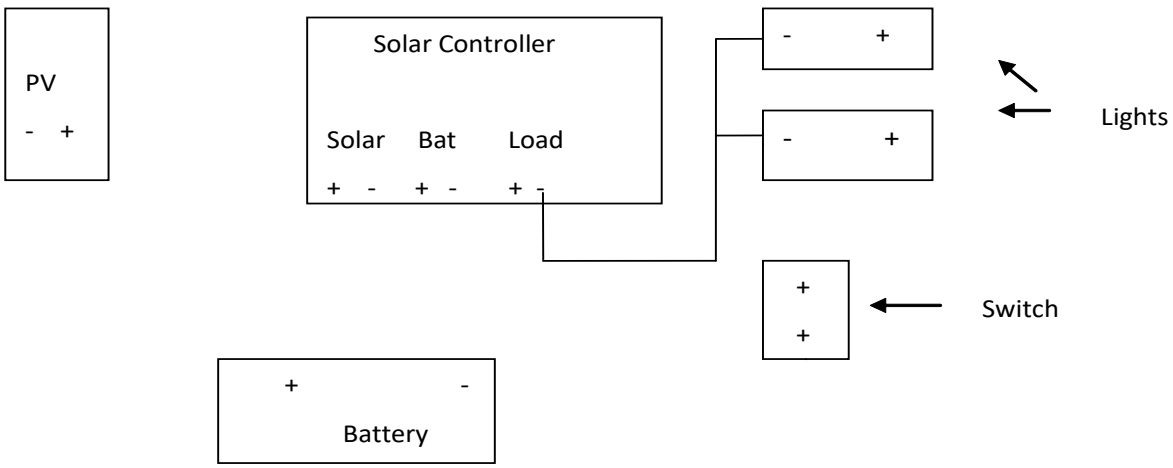


Figure 3: Wiring Diagram of 1 module 2 lights Solar Home System

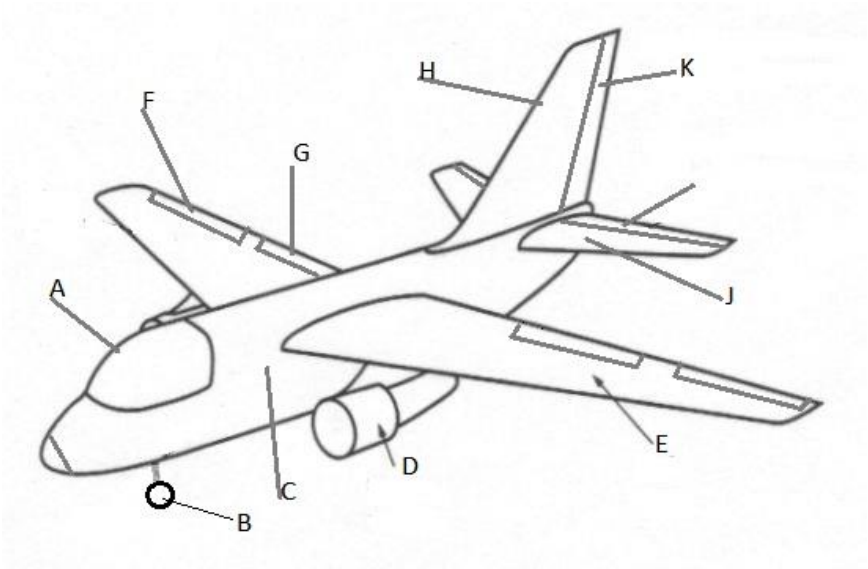


Figure 6

Q20 State the names of the parts of figure 6 as indicated in the table below ***(10 marks)***

LABEL	***Name***	***LABEL***	***Name***
A		*F*	
B		*G*	
C		*H*	
D		*J*	
E		*K*	

I. A. Edwin
A.Agyei-Agyemang

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December 2010

ANSWER ALL QUESTIONS IN PARTS I, II & III SEPARATELY.
NO BOOKS ARE ALLOWED
PART I (MECHANICAL)

1.a. Write the appropriate name of the machine tool shown in Figure 1 and complete the label of the parts indicated. (15 marks)

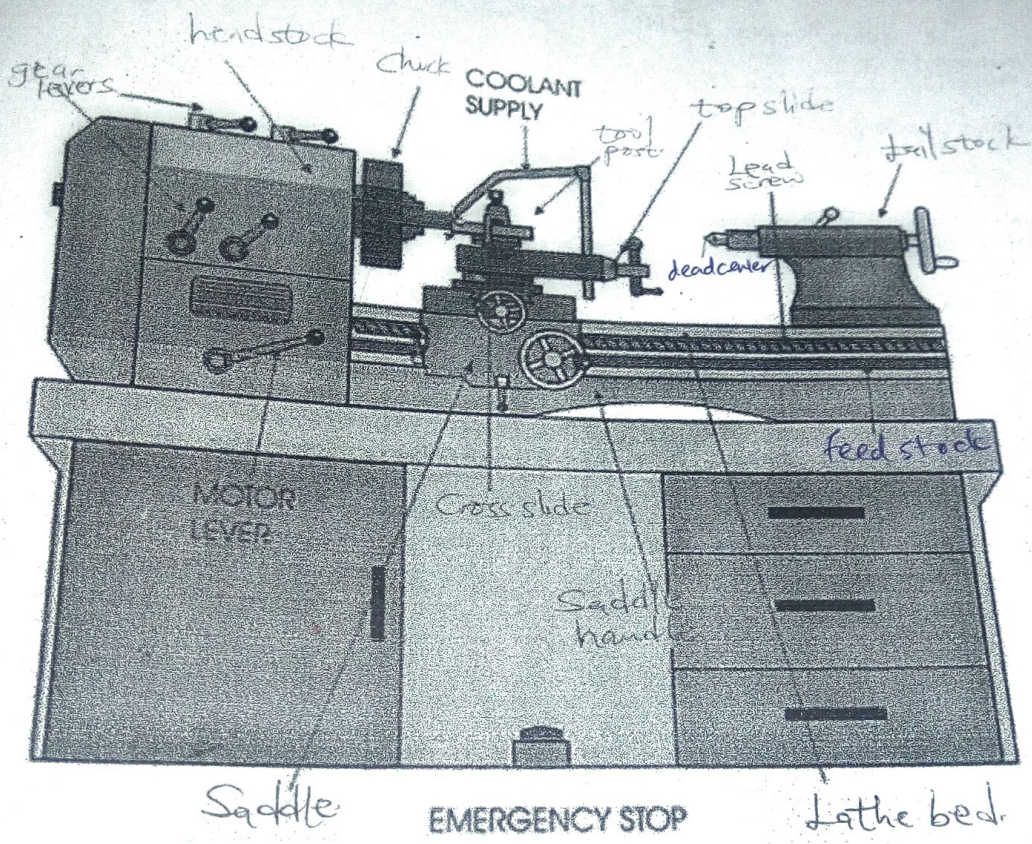


Figure 1: Center Lathe machine

b. Briefly state the functions of any eight parts as labelled in figure 1
(8 marks)

i. Jaw chuck holds the piece of work to be worked on
ii. Tool post enable the tool to be fixed into it holds the tool.
iii.
iv.
v.
vi. The gear levers control the speed of the chuck.
vii. The cross slide is used for taking depth of cut when turning.
viii. The compound slide is used to take the depth of cut when facing
- The lead screw or shaft is used to cutt screw threads
- The feed rod or shaft is used for smooth turning.

2.a . Write the appropriate name of the machine tool shown in Figure 2 and complete the parts indicated

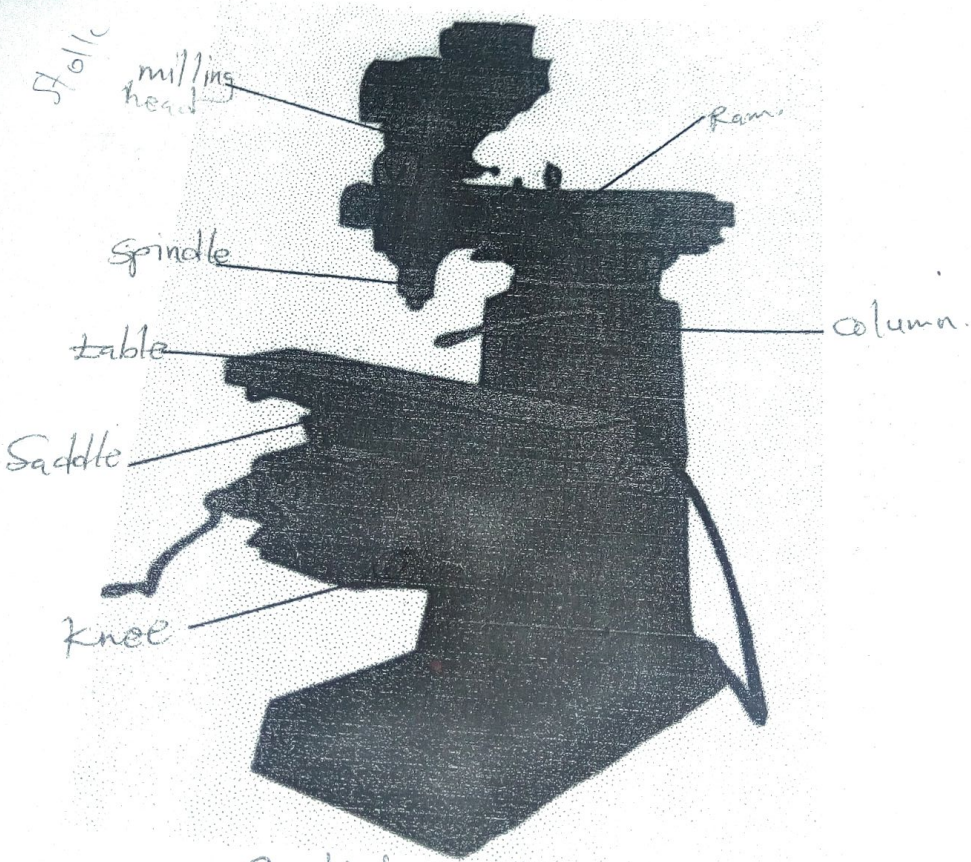


Figure 2: Radial Drilling machine

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b. Briefly state the function of any four parts as labelled in figure 2
(4marks)

i.

ii.

iii.

iv.

3. Write down the names of the parts A-E indicated on the steam boiler in Figure 3 and state the functions of each one (15 marks)

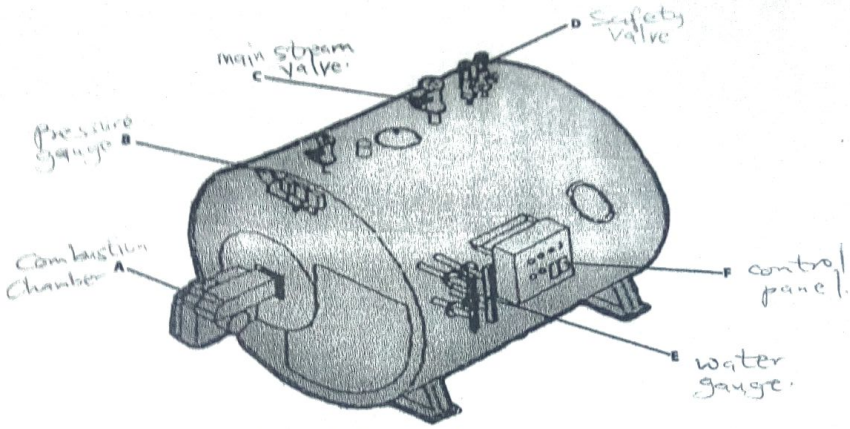


Figure 3: Steam boiler

LABEL	FUNCTION
A. Combustion Chamber	It helps in combustion of the fuel.
B. Pressure gauge	It shows the amount of steam in the Boiler
C. Main stream valve	It controls the pressure of the steam
D. Safety valve.	It prevents explosion.
E. Water Gauge.	It indicates the amount of water in the boi
F Control panel	It controls

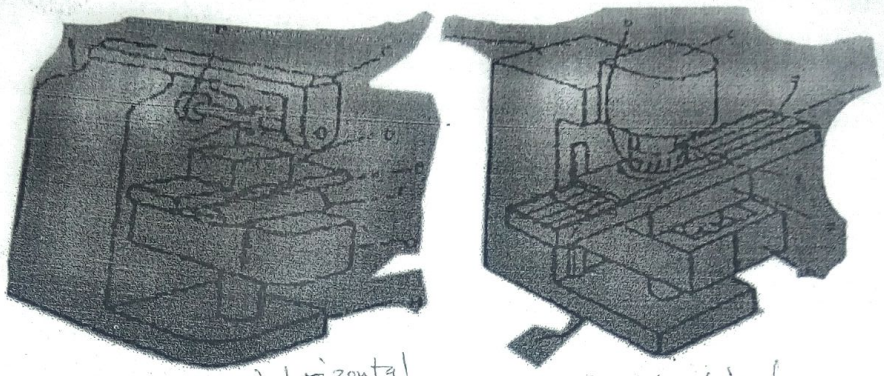
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4. Write down the appropriate names of the two milling machines shown in Figure 4 and 5 (5 marks)



3) horizontal milling machine

4) Vertical milling machine

5. What machine tool is best suited for producing each of the following ? (12 marks)

a. gear teeth..... Universal milling machine.....

b. external keyway ..... Universal milling machine.....

c. internal keyway ..... Universal milling machine.....

d. multiple holes in a flat plate

..... Radial drilling machine. .....

.....

e. smooth flat surface ..... Surface grinding machine.....

.....

f. thread on a short shaft ..... Center lathe machine.....

.....

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Did you know that citations can make your loved ones smile?

6. Suggest four parts that are common to air compressors and automobile engines (8 marks)

i. Piston ii. Valves

iii. Cylinder head iv. Piston rings.

7. Suggest four uses of compressed air (4 marks)

i. for inflating tyre. ii. for hoisting purposes

iii. for spraying purposes iv. for braking mechanisms in big trucks

8. Suggest four differences between a Diesel Engine and a Petrol Engine (4 marks)

	Petrol	Diesel
i.	It uses petrol as fuel	It uses diesel as fuel
ii.	It uses spark plug	It does not use spark plug
iii.	Power stroke is required	Power stroke is not required
iv.	Fuel and air are drawn into the cylinder	Only air is drawn into the cylinder
v.	Has carburetor	donot have carburetor.
vi.	Has contact breaker	donot have a contact breaker

9. State any SIX applications of solar energy (12 marks)

i. Solar energy is used to power traffic lights

ii. It is used to power communication systems

iii. It is used to power satellites.

iv. It is used to power street lights.

v. It is used to power calculators.

vi. It is used to power industrial steam boilers.

10a. Sketch and label a carburizing flame in the box (3 marks)

b. A carburizing flame comprises more acetylene gas and less oxygen gas (2marks)

11. An oxidizing flame comprises more oxygen and less acetylene gas

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12. It is required to provide 220 - 240 volts solar power system to one of the classrooms in the college. Make a list of six components and materials required. Do not give quantities and state the purpose of each one. (12 marks)

Component/ material	Purpose
i. Inverter	Convert Dc to AC
ii. Battery	Supply and receive energy
iii. Regulator	it controls the energy discharge of the battery
iv. Solar panel	It is used to trap solar energy from sun
v. Switch	It is used to open and close the circuit
vi. load [bulb]	electrical consuming component.

13. Name any three alternate sources on non-conventional energy (6 marks)

i. Tidal energy
ii. Solar energy
iii. Hydro-electric energy
iv. Wind energy
v. Biomas
vi. Geothermal energy

14. List any five types of fire extinguishers (5 marks)

i. Carbon tetrachloride fire extinguisher
ii. Carbon dioxide ✓
iii. Large foam ✓
iv. Water ✓
v. Dry chemical ✓

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15. State one function of each of the following in the automobile (32 marks)

a. ignition coil ... It converts low voltage to high voltage to create a spark in the spark plug

b. Condenser ... It absorbs the electric current thereby preventing interference of the contact points

c. crankshaft ... It converts the reciprocating linear motion of the piston into rotational motion.

d. gudgeon pin ... It connects the piston to the connecting rod and provides a bearing for connecting rod to pivot upon as the piston moves

e. Radiator ... It helps to remove excess heat from the engine

f. Carburetor ... It blends fuel and air for internal combustion.

g. Piston rings ... It seals the combustion chamber to prevent transfer of gases.

h. Piston ... It supports heat transfer from the piston to the cylinder wall.

I. Flywheel ... It stores kinetic energy and keeps revolving inertia of the engine through its cycle

j. Alternator ... It keeps the battery charge to provide power for the vehicle

k, Battery ... It supplies electric power to move the vehicle

l Gearbox ... It transmits power in the system.

m. Propeller shaft ... It rotates to move the vehicle

n. Clutch ... It engages or disengages the power of the engine from rest of transmission.

o. Contact Set ... It interrupts current flowing in the primary circuit of the ignition coil It opens to collapse the magnetic field in the coil thereby inducing a high voltage through the secondary winding to spark plug.

p. Cam shaft ... It is used in closing and opening of the fuel valves.

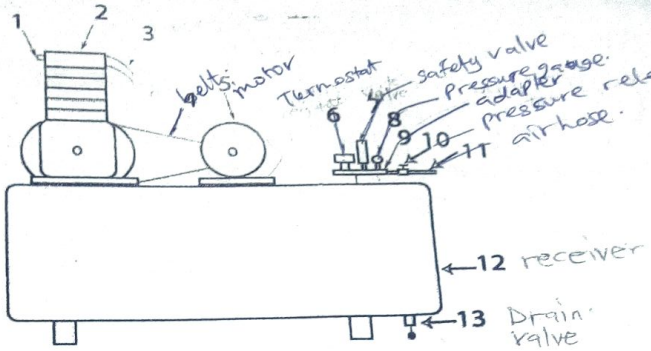
16. LABEL

17. a. L (11 ma

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16. LABEL THE DIAGRAM SHOWN BELOW. What equipment is this?



17. a. LABEL THE DIAGRAM SHOWN BELOW (figure 6)
(11 marks)

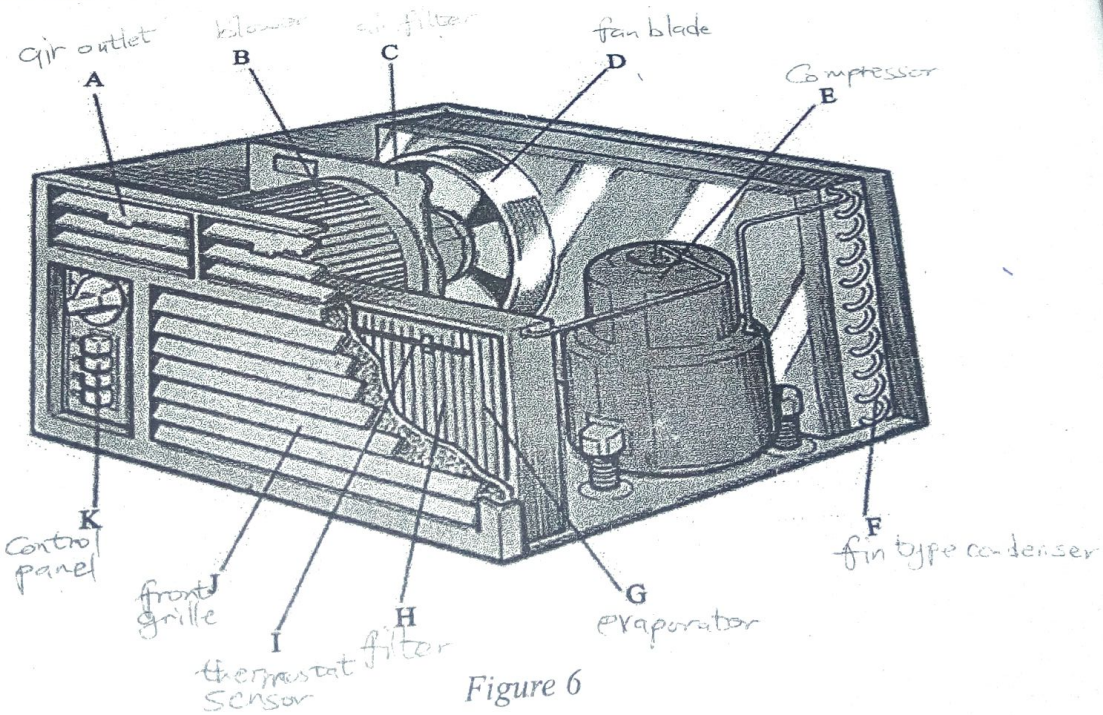


Figure 6

Do you want a citation that will not break?

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A. Air outlet
B. Blower
C. air filter
D. Fan Blade
E. Compressor
F. Fin type condenser.
G. Evaperator
H. filter
I. Thermostat sensor
J. Front grille
K. Control panel

b. what are the functions of : (10 marks)

B. It blows the heat in the evaporator coil air away
D. It pulls outside air through the cabinet outside vents and blows it over the condenser coils
E. It compresses the low pressure refrigerant into high pressure gas.
F. It condenses the high pressure gas into liquid and gives off heat to the outside air
G. It absorbs the heat from the inside air and converts it from liquid to gas

18. With the aid of a diagram in the box below describe how a basic refrigerator works (10 marks)

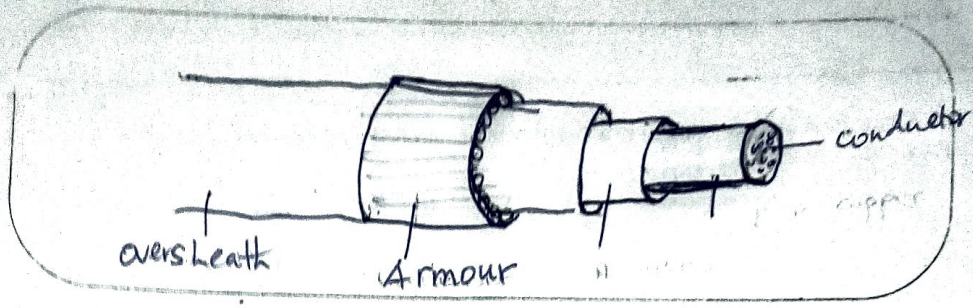
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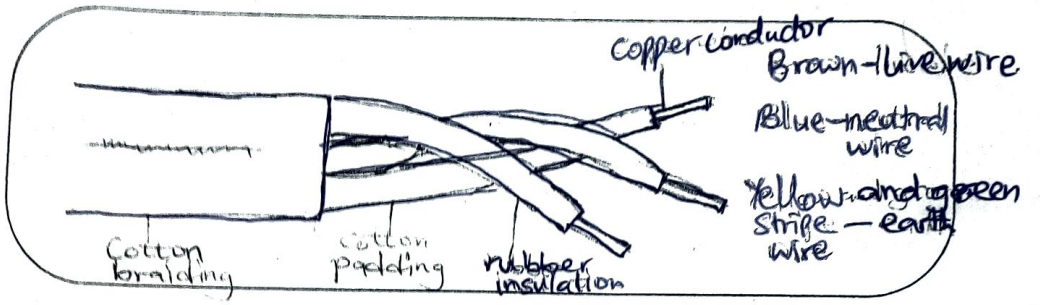
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19. Sketch and fully label a single- core PVC - insulated wire armoured PVC -covered (PVC/SWA/PVC) cable. (5 marks)



20. Draw a three core flexible cable and fully label it in the space provided stating the colour codes (5 marks)



21. a. Explain the operation of the following switches: (4 marks)

Single Pole ~~Doubel~~ Throw switch

Double

It has 3 terminals. The middle terminal is connected to one of the supply l
So when the switch is flipped to one side two of the three terminals come in
contact and when the same switch is flipped to the other side the other
two combination also come into contact with each other. It controls only one circ

Double Pole Single Throw switch

It controls ~~two~~ independent circuits and act as two identical switches th
are linked mechanically. Both terminals are ~~be~~energize at the same time
They are independent of eachother. ~~and control~~

b. List four types of electrical switches in the spaces provided.

transmission

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ii. .....

iii. .....

iv. .....

22. Draw the wiring diagram of two **lamps** being controlled by 2 switches in the space provided (5 marks)

23. Label the diagram in Figure 7 for an oxy-acetelyn cutting system. (8 marks)

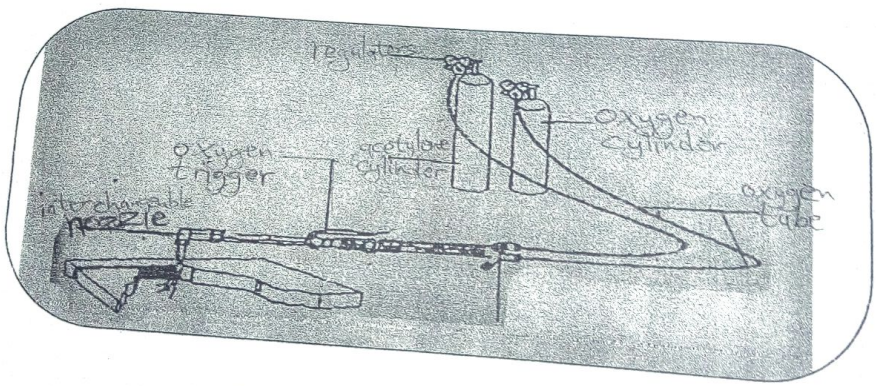


Figure 7 Gas welding set

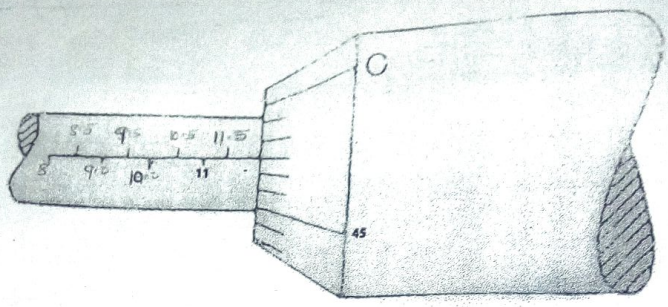
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b. State the reading on the micrometer shown in figure 9 (3 marks)



Ans: . 11·5 + 0·45 + 0·03 = 11·98 mm

c. State the instrument you would use to measure the following (7 marks)

I. An internal measurement (bore) to an accuracy of two decimal places
Vernier calipers

ii. An internal measurement (bore) to an accuracy of three decimal places
Vernier Calipers

iii. The depth of a hole estimated to be about 51mm to two decimal places
Depth micrometer screw gauge.

iv. The depth of a hole estimated to be about 51mm to three decimal places
Internal micrometer calliper

v. A round solid bar 55.355mm in diameter External micrometer screw gauge.

vi. The included angle of a taper gauge. Universal bevel protractor

vii. Checking a horizontal surface for leveling Engineering Steel rule.

Edwin, I. A
Idan, C. K

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A..... B..... C..... D..... E..... F..... G..... H..... I.....

J..... K..... L..... M..... N..... O..... P..... Q..... R.....

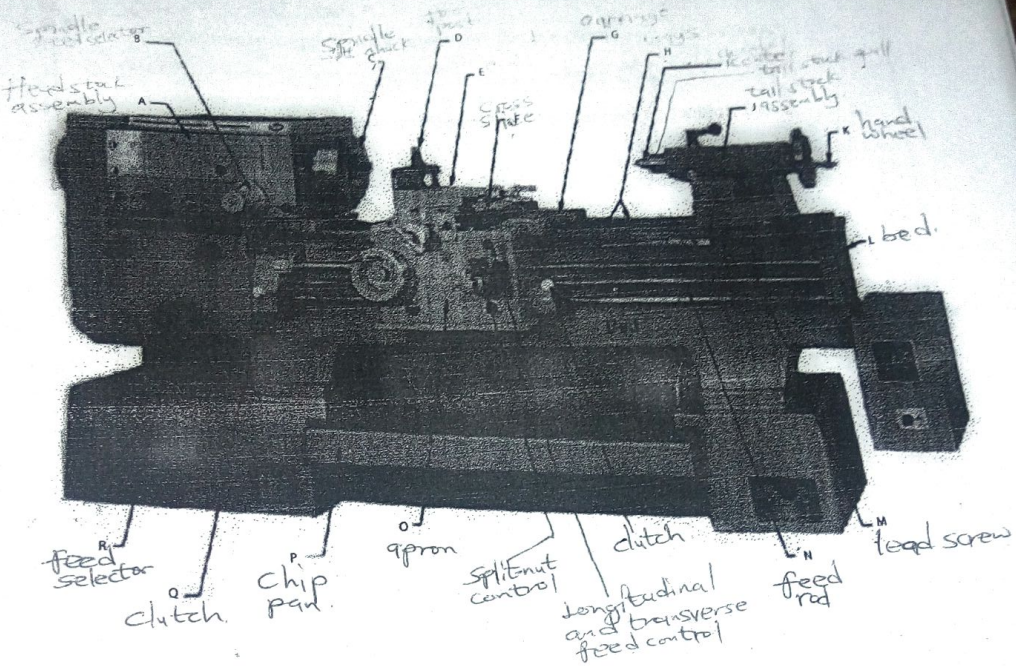


Fig. 2 LATHE

3. Write down the appropriate names of the milling machines shown in Figure 3 and 4 and label parts indicated

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4. Fig 5 shows a basic refrigerator cycle. Lable the parts shown A-D

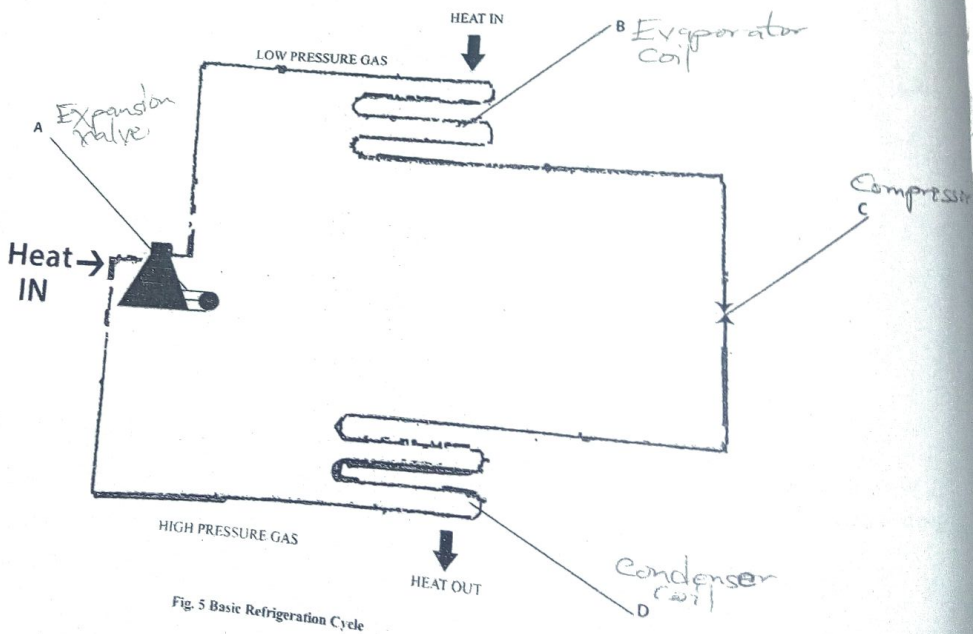


Fig. 5 Basic Refrigeration Cycle

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5. What are the following of the parts labelled A- D?

A. Expansion valve

B. Evaporator coil

C. Compressor

D. condenser coil.

6. For an air conditioner. suggest two components that are missing in Fig 5 above

a. blower, Fan, air filter b. air passage, air outlet.

7. State three main differences between a Radial Drilling Machine and a Drill Press

1)

8. What machine tool is best suited for producing each of the following?

a. gear teeth Universal milling M. b. external keyway Universal milling machine

c. internal key way Universal milling Machanee d. multiple holes in a flat plate Radial drilling machine.

e. Smooth flat surface Surface grinding machine f. one million bolts Universal milling machine.

g. threads on a short shaft Center lathe machine.

9. Insert the missing words in the follwoing

Generally common law duty can be divided into four specific groups as follows:

1. The duty to provide Safe place of work

2.The duty to provide proper plant machinery and equipment.

3.The duty to organise and maintain a safe system of work

4. The duty to employ competent (skilled) workers.

10. Suggest four parts that are common to air compressors and automobile engines

~~Expansion valve~~ Piston. - Valves. -rings

Compressor Cylinder head. - Connecting rod.

and suggest four uses of compressed air

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11. List six causes of accidents

2) Unsafe personal conditions.
3) Unsafe acts
4) Irregular inspection of equipment.
5) Unsafe conditions of the machine or equipment.
6) Poor maintenance of equipment

12. Name any three types of fire extinguishers

1) Dry chemical fire extinguisher.
2) Large foam fire extinguisher.
3) ~~Water fire extinguisher.~~
4) Carbon tetrachloride fire extinguisher
5) ~~Carbon~~ dioxide ~~fire extinguisher~~

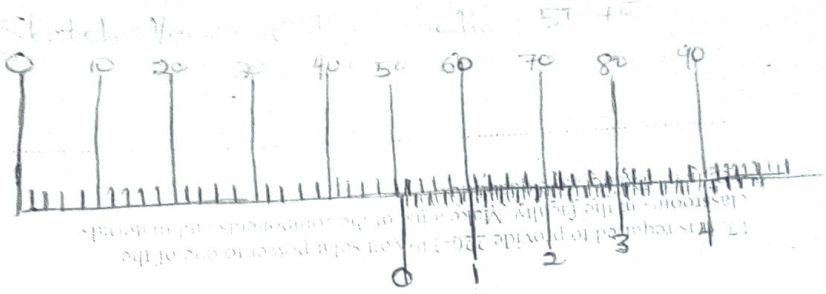
13. Suggest any four ways in which occupational diseases from dust and fumes cab be prevented.

1). Wearing personal respiratory equipment
2) Ensure natural ventilation.
3) Local ventilation of the source of the contaminant
4) Wearing contaminant control equipment.

14. List four advantages of automobile sprinklers for fire fighting

1) They are always ready to start operation.
2) They prevent waste of water, they only activate ~~t~~when needed.
3) They operate where fire fighters cannot enter.
4) They begin to operate even before the fire starts and prevents the fire from ~~be~~becoming dangerous.

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18. State any six application of Solar energy

Used to power satellite.
" " " streetlights.
" " " calculator
" " " traffic light.
" " " communication system.
" " " industrial steams boilers.

19. Sketch an automobile chassis and label any four parts

20. State one function for each of the following in the automobile

a. ignition coil ...It transforms low voltage to high voltage to create a spark in the spark plug.

b. condenser...It absorbs the current thereby preventing interference at the contact points.

c. crankshaft...It converts the reciprocating linear motion of the piston into rotary motion.

d. flywheel...It

e. alternator...It keeps the battery charged to provide power for the vehicle

f. battery...It supplies electric power to move the vehicle.

g.carburetor...It blends air and fuel for internal combustion.

h.piston...It supports heat transfer from the piston to the cylinder

I.radiator...It helps to remove excess heat from the engine.

j. Pistion rings...It seals the combustion chamber to prevent the transfer of g

21. Suggest four differences between a diesel engine and a petrol engine

Petrol engine	Diesel engine
- It uses spark plug.	It does not use spark plug.
- Petrol is the fuel used	Diesel is the fuel used.
- Fuel and air are drawn into the cylinder	Only air is drawn into the cylinder
- Power stroke is required	Power stroke is not required.

22. How would you specify They can be specified according to their functions. cutt Sort this

a. Lather machine - It is used to machine cylindrical parts, drill, holes, bore holes and cut
b. Milling machine - It is used drill holes, cut external splines, cut gears, machine flat surfaces
c. Radial Drilling machine - It is used to drill and bore holes, and enlarge holes
d. shaper - It is used to machine flat surfaces and cutt gear teeth.

/

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and tithe cards

December 2013

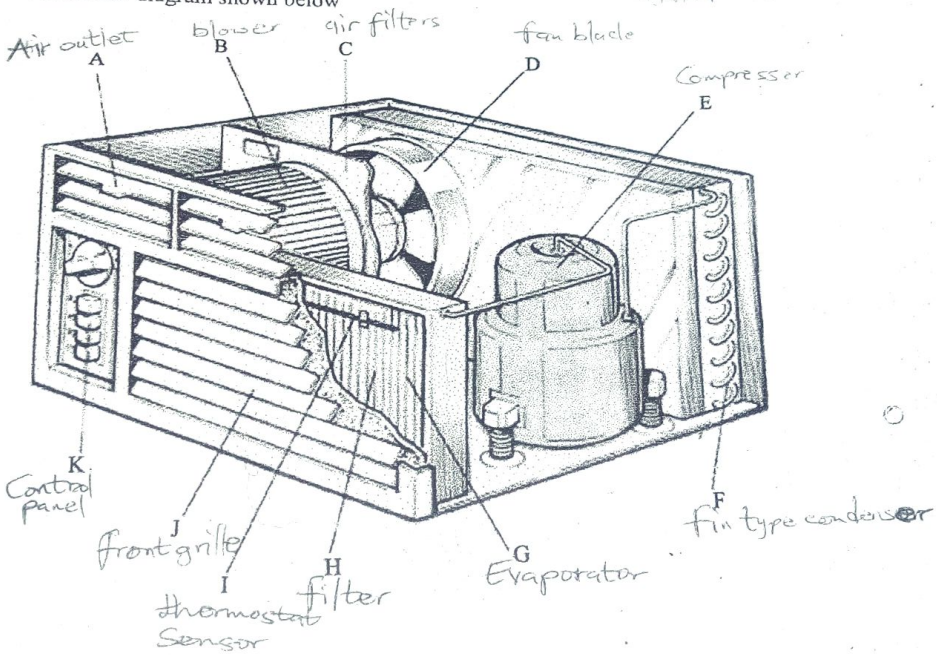
The questions in this section are not arranged in the order in which they appeared in the original question paper

1. Draw and lable the center lathe machine

2. It is required to provide a 220-240 volt solar power system to one of the classrooms in the college. Make a list of six components and materials required. Do not give quantities and state the purpose of each one.

3. Name any three alternate sources of non-conventional energy tidal, geothermal, biomass, solar, hydro, wind

4. Label the diagram shown below



b. what are the functions of B, D, E, F, G,

5. Sketch and fully label a single -core PVC insulated steel wire armoured PVC- covered (PVC/SWA/PVC) cable

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6. a. Label the steam boiler shown in the diagram below

b. Suggest any four uses of steam generated from a steam boiler.

7. Refrigeration follows a cycle. Complete the cycle of refrigeration by putting in the missing (or appropriate) terms.

Nota Bene: this particular question is referenced to the original question meanwhile the exact content could not be obtained verbatim.

8. a. Label the diagram of an air compressor shown in the diagram .

b. Suggest four uses of compressed air

9. Label the diagram of an oxy- acetylene cutting system.

10. What machine tool is best suited for producing each of the following.

a. gear teeth — U. milling machine b. external keyway – Universal milling machine

c. internal keyway U. milling machine d. multiple holes in a flat plate – Radial drilling machine

e. smooth flat surface Surface grinding machine f. one million bolts

g. thread on a short shaft – Center lathe machine

11. Insert the missing words in the following

Generally common law duty can be divided into four specific groups as follows:

1. The duty to provide ..... Safe ..... place of ..... work .....

2. The duty to provide proper plant ..... machinery ..... and ..... equipment .....

3. The duty to ..... organise ..... and maintain a safe system of ..... work .....

4. The duty to employ competent (skilled) ..... workers .....

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KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

B.Sc. (Engineering) First Semester Examination, 2013, First Year

ME 195/CE 153: ENGINEERING TECHNOLOGY **TIME ALLOWED: 3 Hrs**

ANSWER ALL QUESTIONS. In PARTS I, II & III SEPARATELY.

NO BOOKS ARE ALLOWED ***INDEX NUMBER:*** **……**

PART I (MECHANICAL) ***PROGRAMME*/DEPARTMENT: ……..**

Q1. Sketch in 2D in the space below any of the CENTRE LATHE machine in the machine shop and label the following: Tailstock, Headstock, Compound Slide, Tool holder, Cross slide, Feedshaft, Leadscrew, Barrel, Tool post, Chuck, Jaws, Gear levers, Dead Centre, Bed, Apron ***(30 marks)***

Q2. P**arts in an automobile.** ***(10 marks)***

i. State the difference between Crankshaft and Camshaft ……
……..……

ii. State what is common with Crankshaft and Camshaft ……....
……..

iii. What is Dead Center in relation to engine Piston ……

……..

iv. State one function of Battery ……..……..

……

CHOOSE THE CORRECT WORD/SET:

A car uses a four-stroke engine. The four strokes are

intake, compression, ignition and exhaust

injection, rotation, ignition and exhaust

injection, carburetion, rotation and exhaust

Q3. Insert the name of the appropriate machine in the following sentences: ***(15 marks)***

A …… Machine is generally/commonly used to manufacture gear teeth

A ……..……Machine is generally/commonly used to manufacture external keyways

A ……..… Machine is generally/commonly used to manufacture internal keyways

A …… Machine is generally/commonly used to make multiple holes in a flat plate

A …… Machine is generally/commonly used to make smooth flat surfaces

A ……. Machine is generally/commonly used to manufacture one million bolts

A …… Machine is generally/commonly used to manufacture thread on a short shaft

Q4. Fill in the blank spaces with the appropriate word: ***(10 marks)***

In arc welding, the intense …….. needed to melt metal is produced by an …….. arc. The arc is formed between the actual work and an …… that is manually or mechanically guided along the joint. Arc welding may be done with ……. current with the electrode either positive or negative or …… current. The choice of current and polarity depends on the ……, the type of ……, the arc atmosphere, and the ……. being welded

Q5. Fill in the blank spaces with the appropriate word ***(10 marks)***

In a refrigerator, the …….. compresses the ……... gas. This raises the …… pressure and …….., so the heat-exchanging coils outside the refrigerator allow the …….. to dissipate the heat of pressurization.
As it cools, the refrigerant ……..... into liquid form and flows through the …… valve. When it flows through the …….. valve, the liquid refrigerant is allowed to move from a …….. pressure zone to a ……. pressure zone, so it expands and ……... In so doing, it absorbs ……..…..... making it cold. The coils inside the refrigerator allow the …….. to absorb ……, making the inside of the refrigerator cold. The cycle then repeats.

Q6. State the instruments you would use to measure the following: ***(10 marks)***

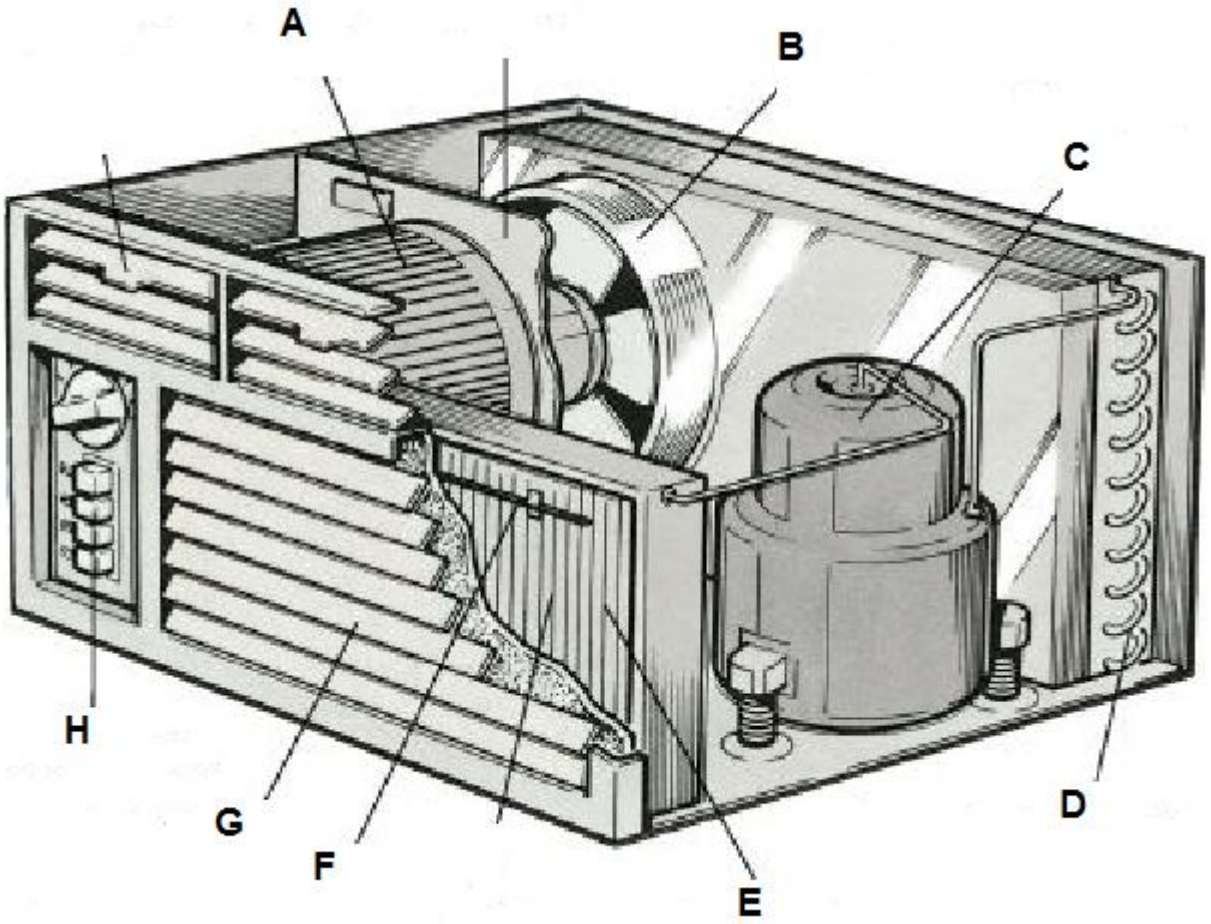
An internal measurement (bore) to an accuracy of two decimal places ……...

The depth of a hole estimated to be about 51 mm to two decimal places ……

A round solid bar 55.35 mm. in diameter ……..

Checking a horizontal surface for leveling……..

Q7. LABEL THE DIAGRAM SHOWN BELOW ***(15 marks)***



A…… B…….

C……. D……..

E……. …… F ……..

G……. H ……...

What are the functions of?

B……...

C…….

D…….

F……..

G……..

Q8. It is required to provide 220-240 volt solar power system to one of the classrooms in the college. Make a list of six components and materials required. Do not give quantities and state the purpose of each one. ***(12marks)***

	Component/material	**Purpose**
i.	……	……..
ii.	……	……..
iii.	……	……..
iv.	……	……..
v.	……	……..
vi.	……	……..

Q9. Name any three other alternate sources of non -conventional energy ***(6 marks)***

i.……

ii.……

iii.……

Q10. Suggest **four** uses of compressed air ***(4 marks)***

i. ……

ii. ……

iii. ……

iv. ……

Q11. Sketch and label a carburizing flame in the box ***(3 marks)***

Q12 Sketch and label an oxidizing flame in the box ***(3 marks)***

Q13 Write down the names of the parts A - E indicated on the steam boiler in Figure 1. And state the function of each one ***(15 marks)***

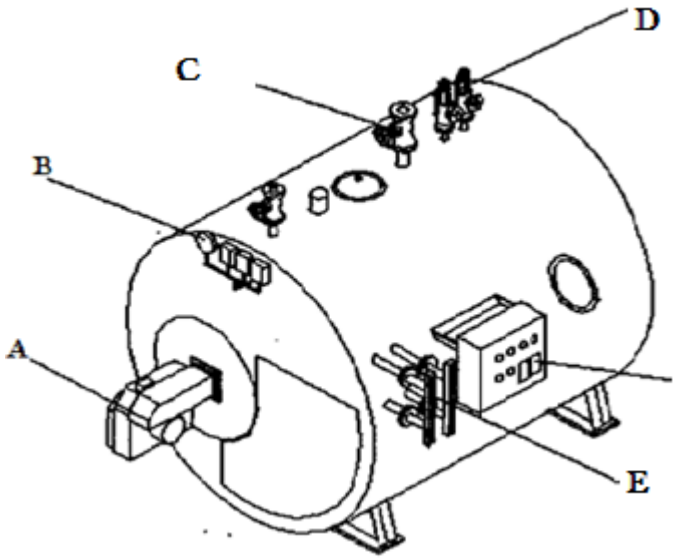


Figure 1: Steam Boiler

LABLE	FUNCTION
A: ……	…….
B. ……	…….
C. ……..……..	……..…….
D…….	…….....
E. ……..	……..……

Q14. Write down the appropriate names of the two milling machines shown in Figures 2 and 3 ***(4marks)***

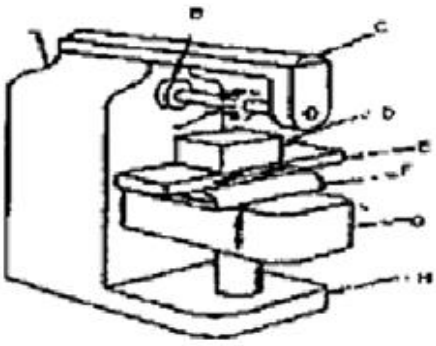


Figure 2: ……..

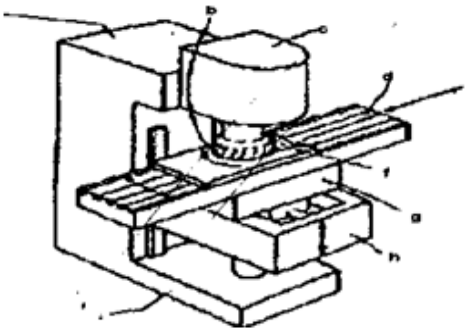


Figure 3: …….

Q15. List any four types of fire extinguishers ***(4 marks)***

i. ……

ii. ……

iii. ……

iv. ……

Q16. Label the diagram in Figure 4 for an oxy-acetylene cutting system. ***(14 marks)***

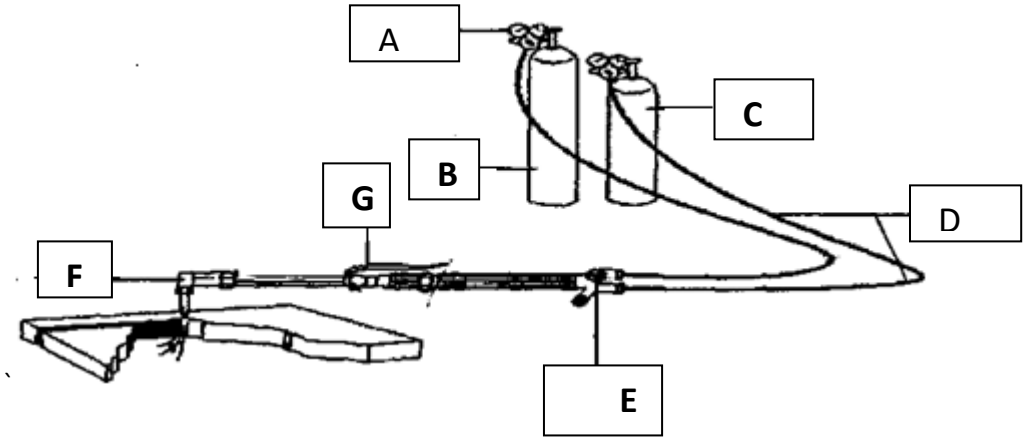


Figure 4: Gas welding set

A …… B ……..

C……... D……..

E …….. F……...

G……..

Q17. State the reading on the micrometer shown in figure 5 *Ans:* ..... ***(3 marks)***

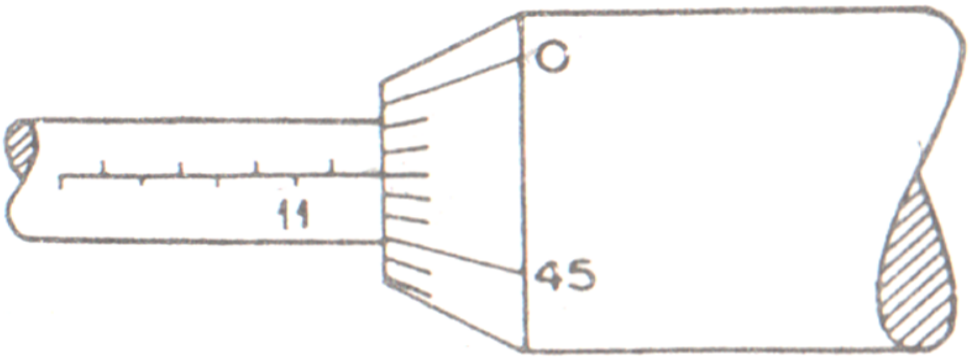


Figure 5

Q18. Sketch and fully label a single-core PVC-insulated steel wire armoured PVC-covered (PVC/SWA/PVC) cable. ***(5 marks)***

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Q19. Explain the operation of the following switches: ***(4 marks)***

i. **Single Pole Double Throw** switch ……..

…..……

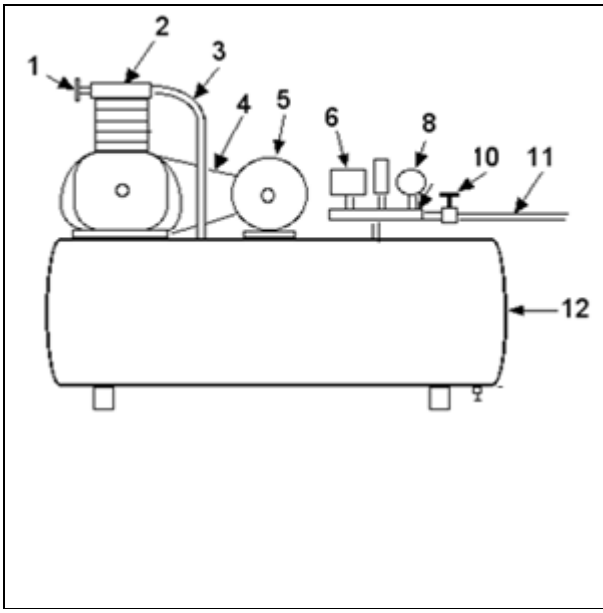
ii. **Double Pole Single Throw** switch ……..

..……

Q20. State the institution of State responsible for the compliance of the Industrial Act **: *(5 marks)***

.....

Q21. LABEL THE DIAGRAM SHOWN BELOW. What equipment is this? .....
(12 marks)



1	
2	
3	
4	
5	
6	
8	
10	
12	

Q22. Suggest **<u>four</u>** uses of compressed air ***(4 marks)***

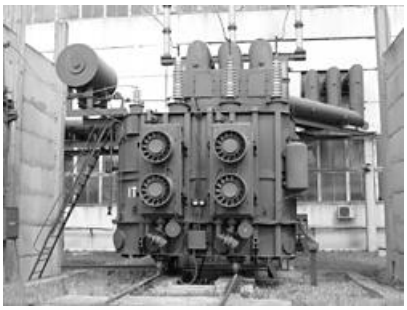
i. …… ii. ……

iii. …… iv. ……

Q23. Electricity generated at power plants has to get to the consumer through cables and other devices. State the names of the electrical devices and equipment listed below. ***(15 marks)***







A …… **B. ……** **C ……**

Q24. List the order in which the following electrical components A-H listed below will be connected for electricity generated to be supplied to a household: ***(10 marks)***

A. Main Switch **B**. Consumer Control Unit (CCU) **C.** Distribution Line **D**. Transformer Sub-Station **E**. Transmission Line **F.** Cut Out **G** Energy Meter **H**. Service Cable

1 _____ *2.* _____

3. _____ *4.* _____

5. _____ *6.* _____

7. _____ *8.* _____

Q25 The peak value of irradiance (solar radiation) at the earth's surface is? ***(2 marks)***

a) 1 kW / m
b) 1 kW / m^2
c) 1 kW / cm^2
d) 1.5 kW / m^2

Q26 In my room we have the following appliances and operate them on daily basis. Calculate my daily energy consumption in Watt-hours (Wh), kilowatt-hours (kWh) and units of electricity (Units).
(5 marks)

No.	Appliance	Power (Watts)	Time (hours)	Energy
1	refrigerator	100	12	
2	water heater	1000	½	
3	Electric Iron	750	½	
4	Rice cooker	900	½	
5	Television	80	6	
6	Standing Fan	70	6	
7	Laptop Computer	60	6	
8	Compact Fluorescent Lamp (CFL)	20	8	
			Total	

i. …… Wh ii. …… kWh . iii. …… Units

Q27 Complete the wiring diagram of a Solar Home System (SHS) figure 6 below by joining the correct connecting terminals. ***(10 marks)***

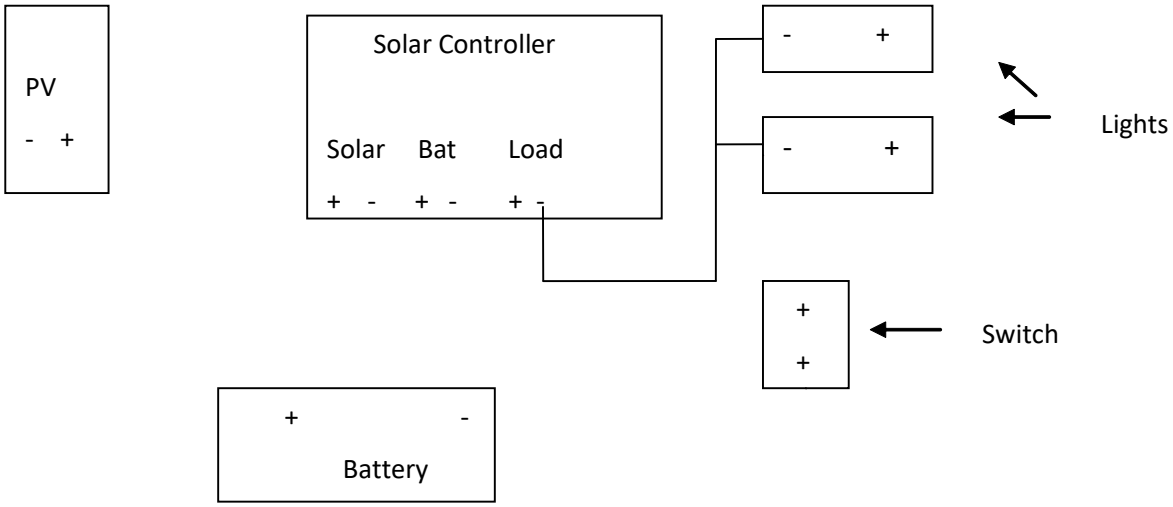


Figure 6: Wiring Diagram of 1 module 2 lights Solar Home System

Q28

I. A. Edwin